

ASTHMA

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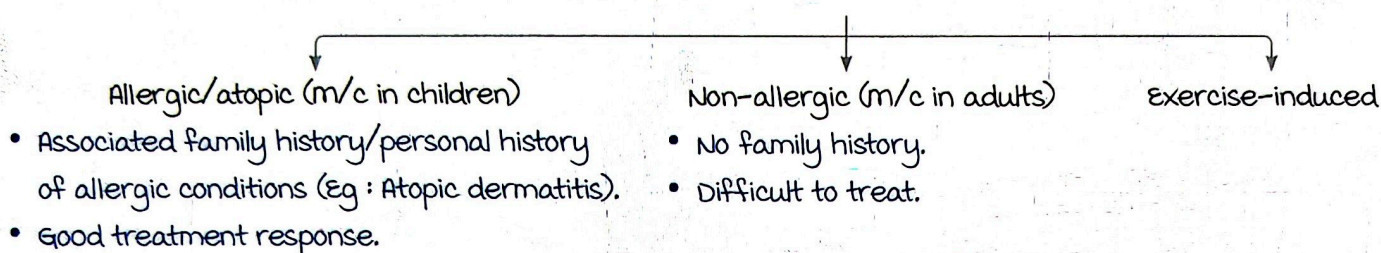
Introduction

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CHARACTERISTICS :

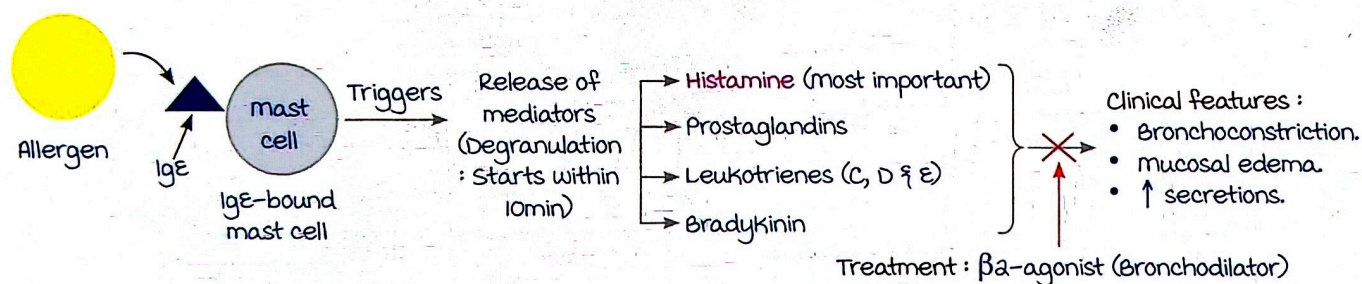
1. Bronchoconstriction d/t → Hyper-responsiveness of airway
(Reversible airway obstruction) → Chronic airway inflammation
2. Variable intensity & frequency of episodes over time : management according to subjective symptoms.

CLINICAL PHENOTYPES :



PHASES OF ALLERGIC ASTHMA :

Early phase :



Late phase :

Starts : 3 to 4h later.

Peak : By 10 to 12h.

Inflammatory cells recruitment (Eosinophils, basophils & lymphocytes) → Repeated exposure → Chronic airway inflammation.

↑(-)
Steroids

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CLINICAL FEATURES :

- Feedback
- Recurrent episodes of wheezing/cough/chest tightness.
 - Varying intensity and frequency (↑ frequency late night/early mornings).

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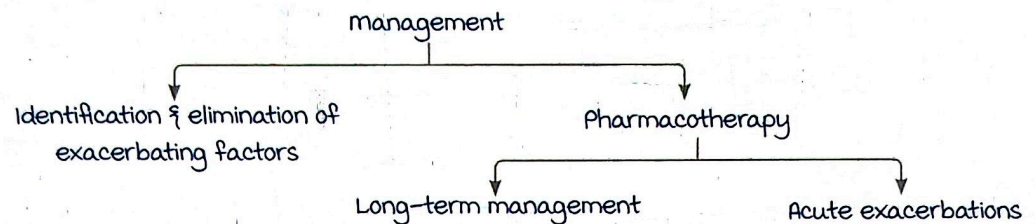
LUNG FUNCTION ABNORMALITIES :

Condition	Parameters
Air-flow limitation	FEV ₁ ↓ FEV ₁ /FVC ratio < 0.8
Bronchodilator response	FEV ₁ ↑ by > 12%
Exercise challenge:	Air-flow obstruction worsened: FEV ₁ ↓ by > 15%
Diurnal Variation/Am to Pm variation	FEV ₁ , or PEF (Peak Expiratory Flow) ↑ by > 20%

Management of Asthma

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Based on global initiative network for asthma (GINA) guidelines.

**LONG TERM MANAGEMENT :****Age >12y :**

managed as in adults.

Age 6 to 11yrs :**Stepwise management.**1. Reliever therapy → Short-acting β -agonist (SABA) (SOS for sudden exacerbation of symptoms)

Frequency of Symptoms	Step of therapy	Controller therapy (Long-term)
< 2 / month	I	Whenever SABA taken → ICS lowdose
> 2 / month; not daily	II	Daily low dose ICS
Daily symptoms + Night symptoms $\geq$ 1/week	III	Daily low dose ICS + LABA OR Daily medium dose ICS
Daily symptoms + Night symptoms $\geq$ 1/wk + Low lung function	IV	Daily medium dose ICS + LABA OR, Highdose ICS
uncontrolled on Step IV	V	Daily high dose ICS + LABA (+/-), Biological agents (Last resort)

- SABA : Short acting β -agonist.
- LABA : Long acting β -agonist (Formoterol).
 - Synergistic effect with steroid.
 - steroid sparing effect (Long-term).
- ICS : Inhaled corticosteroid Budesonide used.
 - Low dose : 200 to 400 mcg.
 - medium dose : 400 to 800 mcg.
 - High dose : >800 mcg.
- Biological agents : Immunomodulatory monoclonal antibodies.
 - Omalizumab : Anti-IgE.
 - mepolizumab : Anti-IL5.
 - Dupilumab : Anti-IL4 receptor.
 - Anti-TSLP (Thymic stromal lymphopoietin) : New drug.

Age < 5y (Preschool):

Asthma may resolve >5y in some patients.

Classification of patient: Episodic wheeze.

1. **Episodic wheeze**: During viral infection.

2. **multitrigger wheeze**: During respiratory infection + Other triggers: Cold, exertion, Food.

Asthma probability:

↑ probability for future asthma (Requires treatment).

- Symptoms last >10d after URI And,
- > 3 episodes.

Treatment:

Controller: **Daily low-dose ICS therapy**
+
Reliever therapy: **SABA**

Review after
2 months

Improvement → Stop treatment

No Improvement → Treat as per Age 6 to 11 guidelines.

ACUTE EXACERBATION AND ITS MANAGEMENT:

Classification of exacerbation episodes:

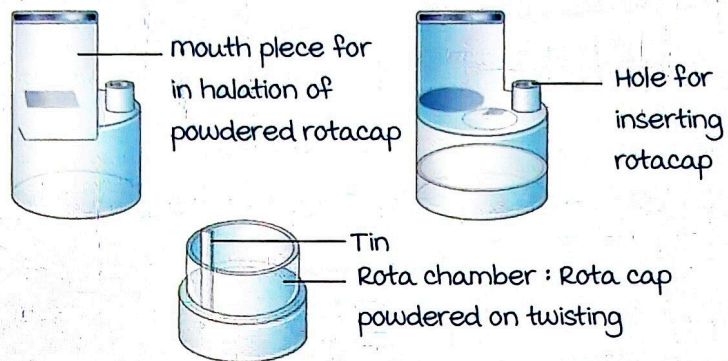
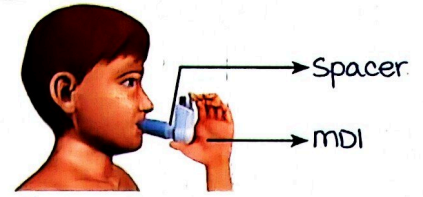
	mild	moderate	Severe
Sensorium	Normal	Anxious	Agitated
Dyspneal retractions	Absent	moderately present	Severe
SpO ₂	>95%	90-95%	<90%
PEFR (% of predicted)	>80%	60-80%	<60%
management	• Salbutamol in metered dose inhalers (MDI) ↓ 4-10 puff every 20 minutes.	• Salbutamol + Ipratropium (Centrally acting anti-cholinergics) • O₂ to keep SpO₂ >95% • Oral prednisolone	• Neb. Salbutamol + Ipratropium • Oral prednisolone or iv Hydrocortisone • Inj magnesium sulphate

PEFR: Peak expiratory flow rate.

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ASTHMA MEDICATION DEVICES :

- Based on age of child :
 - 12 yr : MDI.
 - 4 to 12 yr : MDI + Spacer (Allows drug to be suspended in air → Inhaled over few breaths).
 - < 4 yr : MDI + spacer + face mask.
- Rotahaler (Not commonly used).



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